Phonon mode–coupling comparison: DFT vs MLIPs (CBVN)

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Overview

This script compares the phonon coupling between a variety of instances of MLIPs to the gold standard DFT. for a single defect supercell.

Inputs:

- `CONTCAR_GS` and `CONTCAR_ES` (same atom ordering).
- One DFT `band.yaml` and a set of MLIP `band.yaml` files (Phonopy format).

Output: a text report that (i) selects DFT "coupling" modes from the GS→ES displacement, (ii) scores each MLIP on several diagnostics, and (iii) produces a ranked list.

What the script computes

(1) GS→ES mass-weighted displacement. From the two CONTCARs we compute the minimum-image displacement in Cartesian coordinates, optionally remove the mass-weighted center-of-mass shift, and form the mass-weighted displacement vector:

$$\Delta\mathbf{r}_i = \mathbf{r}_i^{\mathrm{ES}} - \mathbf{r}_i^{\mathrm{GS}}, \qquad \Delta\mathbf{q} = \left(\sqrt{m_1}\Delta\mathbf{r}_1, \ldots, \sqrt{m_N}\Delta\mathbf{r}_N\right) \in \mathbb{R}^{3N}.$$

(2) Coupling-mode weights from DFT eigenvectors. Let $E(\mathbf{q}) \in \mathbb{C}^{3N\times 3N}$ be the DFT eigenvector matrix at a chosen $\mathbf{q}$ (columns normalized). Define per-mode amplitudes and projection power:

$$a_m = \mathbf{e}_m^\dagger \Delta\mathbf{q}, \qquad p_m = \frac{|a_m|^2}{\|\Delta\mathbf{q}\|^2}.$$

The "coupling modes" are the smallest set of modes whose cumulative p_m exceeds the threshold τ.

For MLIP ranking we also define DFT per-mode weights

$$w_m = \begin{cases} |a_m|^2, & \text{kind} = \texttt{p} \\ |\omega_m|\,|a_m|^2, & \text{kind} = \texttt{S} \\ |\omega_m|^2\,|a_m|^2, & \text{kind} = \texttt{lambda} \end{cases}$$

where ω_m is the DFT mode frequency. These weights decide which DFT modes "matter" most when scoring an MLIP.

(3) GS→ES score For a matched DFT/MLIP **q**-point, form the overlap matrix between eigenvector bases:

$$O_{ij} = \left|\mathbf{e}_i^\dagger \tilde{\mathbf{e}}_j\right|^2 \in [0,1].$$

We compute the one-to-one assignment π that maximizes $\sum_i O_{i,\pi(i)}$ (Hungarian algorithm). Using the DFT-defined weights w_m (restricted to modes with $w_m > 0$), we compute:

$$E_{\text{freq}} = \sqrt{\frac{\sum_{m\in\mathcal{V}} w_m \left(\tilde{\omega}_{\pi(m)} - \omega_m\right)^2}{\sum_{m\in\mathcal{V}} w_m}}, \qquad E_{\text{vec}} = \sqrt{\frac{\sum_{m\in\mathcal{V}} w_m \left(1 - O_{m,\pi(m)}\right)}{\sum_{m\in\mathcal{V}} w_m}},$$

and the combined score

$$\text{Score} = E_{\text{freq}} + \alpha\, E_{\text{vec}}.$$

Lower Score is better. I thought this would answer the question: 'which MLIP best matches the DFT coupling modes eigenvalues and eigenvectors, and how well does it do that?' The different kinds of weights allow for different phrasings, for `kind=p` then the question is the above (where coupling modes refer to the vibrational modes containing the largest proportion of the mass weighted displacement vector). for `kind=lambda` the question is 'How well does the MLIP reproduce the DFT modes that dominate the harmonic energetic cost of the GS to ES displacement?' and for `kind=S` the question is a mix. I need to confirm but I think it would be best suited for evaluating how accurate the PL spectra would be.

(4) Secondary diagnostics (subspace agreement). The report also prints:

- *AvgProjPowX*: compares the subspace spanned by the selected DFT coupling modes against the subspace spanned by the selected ML coupling modes, using principal angles. If Q_d and Q_m are orthonormal bases, with singular values σ_ℓ of $Q_d^\dagger Q_m$:

  $$X = \frac{1}{k_d}\sum_\ell \sigma_\ell^2, \quad (0 \le X \le 1).$$

  I thought this score would answer the question: 'How well do the strongly coupled phonons of <MLIP> match those of DFT?'

- *Coupling clusters*: clusters modes by frequency proximity (gap $> \Delta f_{\text{cluster}}$ starts a new cluster), assigns each DFT cluster a weight (fraction of $\Delta\mathbf{q}$ captured), and checks whether ML frequencies in a window around each DFT cluster recover a similar subspace/weight. i thought this would answer the question 'How well does the eigenspace of an <MLIP> frequency range match the displacements predicted by DFT in the same range?'